

Exploration of stacked rotor designs for aerodynamics in hover[☆]Yoonpyo Hong^{a,1}, Dawoon Lee^{a,2}, Sunwoong Yang^{a,2}, Hyojin Kook^{b,3}, Kwanjung Yee^{a,*,4}^a Seoul National University, Seoul 08826, Republic of Korea^b Samsung Electronics Co., Ltd., Gyeonggi 18448, Republic of Korea

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ABSTRACT

The development of rotorcraft with multiple propulsors has increased interest in co-rotating coaxial rotors, also known as stacked rotors, which can generate high thrust while maintaining a compact aircraft design. By optimizing the pitch angle and the blade arrangement, stacked rotors can reduce aeroacoustic noise and vibration while improving aerodynamic performance. This study leveraged a high-fidelity numerical solver to analyze the aerodynamic performance and underlying physics of various stacked rotor configurations in hovering conditions. With three rotor design variables – index angle, stacked distance, and pitch angle difference – together with high-resolution simulations, two key physical phenomena were identified: the inflow effect and the blade-vortex interaction effect. Specifically, the blade-vortex interaction at the lower blade was found to play a critical role in improving hover performance by generating upwash, which increased the local power loading of the lower blade. Also, the unsteady simulations revealed that stacked rotors produced significantly lower peak-to-peak thrust values compared to counter-rotating coaxial rotors, indicating potential reductions in aeroacoustic noise level. Finally, a design optimization based on deep neural networks was performed to successfully extract design rules to take full advantage of inflow and blade-vortex interaction effects, which can be used as a guideline for the future design study of stacked rotors.

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1. Introduction

Recent advancements in distributed electric propulsion have led to the development of advanced rotorcraft configurations, especially for urban air mobility (UAM). UAM aircraft, unlike conventional vertical takeoff and landing aircraft, employ multiple propulsors ranging from four to eighteen [1]. By distributing the required thrust, these multiple propulsors enhance the aircraft's controllability and robustness compared to the conventional single propulsor configuration. Furthermore, when the thrust required by one propulsor is reduced, fluctuations in airloads are reduced. In specific configurations, the blade tip speed of the propulsor can also be decreased. These features result in decreased aeroacoustic

noise, which leads to a quieter flight than conventional helicopters. In addition, noteworthy is the anti-torque system of these multiple propulsors, which allows its torques to be balanced by another propulsor rotating in the opposite direction, thereby eliminating any performance degradation due to the anti-torque system [2].

Coaxial rotors are being considered as an alternative to maintain airframe compactness while increasing the thrust. Coaxial rotors rotating in the same direction, also known as stacked rotors, are one of the main research topics because they can simultaneously improve aerodynamic efficiency and minimize acoustic noise. Prior to the development of UAM aircraft with multiple propellers, stacked rotors were used as scissor-shaped tail rotors [3] or stacked proprotors [4,5]. Notable is the tail rotor of AH-64, which has been demonstrated to improve mechanical clearance and reduce broadband noise.

Over the past ten years, research has been extensively carried out regarding the implementation of stacked rotors on small unmanned aerial vehicles (UAVs) and aircraft designed for UAM. The aerodynamic and aeroacoustic performance of the rotor were studied through experiments and numerical analysis. The effects of various design variables, including index angle, stacked distance, rotation speed, and pitch angle, were examined.

Siena K. S. Whiteside et al. [6] demonstrated that optimized stacked rotors in a UAV can reduce broadband noise and increase

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power loading compared to a single rotor with the same solidity, based on experiments and numerical analysis. Charles E. Tinney and John Valdez [7] showed through experiments that the noise and thrust performance of a stacked rotor at the same speed and pitch angle can vary significantly depending on the different configurations of the stacked rotor. In a subsequent study examining vortex trajectories using particle image velocimetry and digital schlieren, they found that the highest thrust was attained when the lower blade was close to the tip vortex of the upper rotor [8]. Daiju Uehara et al. [9–11] analyzed the aerodynamic performance of the counter-rotating and co-rotating coaxial rotors using experiments. They found that index angle is a key parameter for stacked rotor performance. Subsequently, George Jacobellis et al. [12] used the same experimental data to analyze the effects of airfoil interaction using RCAS and computational fluid dynamics (CFD), and performed a physical analysis of the vortex trajectory. More recently, Chloe Johnson et al. [11] conducted an experiment with pitch angle as a variable and identified that aerodynamic performance and noise levels changed as the pitch angle varied. Rafael Caracuel Rubio et al. [13] and Stefan Platzer et al. [14] analyzed the flowfield for a particular stacked rotor configuration with an extremely dense grid system.

The previous studies above revealed that stacked rotors can outperform a single rotor of the same solidity in terms of aerodynamic performance in hover conditions. It is noted that the index angle should be optimized as it is crucial in hover performance. The stacked rotor has also been shown to exhibit reduced broadband noise in certain configurations [6]. The design spaces for the research above are, however, still limited. For instance, they only took into account conditions for short stacked distances between the upper and lower blades or confirmed the physics of a few specific cases. Considering the fact that UAM aircraft are being developed in a variety of concepts, such as lift plus cruise, vectored thrust, and wingless, a wider design space should be scrutinized. From this context, further studies are needed to confirm the potential of stacked rotors throughout much wider design spaces; as well as to understand the underlying physics and design implications.

The present research explores a spacious design space to delve into the aerodynamic performance of the stacked rotors in hover. Numerical analyses have been conducted for the wide investigation of stacked rotor configurations. The analysis of noise performances was not performed in this study, as a large amount of computing power was required for accurate noise analyses. Both the index angle and the pitch angle difference between the upper and lower blades have been taken into account throughout the design space. Also, the stacked distance between the upper and lower blades is considered an additional design variable. A high-fidelity aerodynamic solver, which provides high accuracy even with a coarse grid system by applying an advanced numerical method specialized for the rotorcraft flowfield, has been adopted for the effective and thorough examination of the aerodynamic performance. The aerodynamic solver can accurately identify phenomena that are difficult to observe experimentally, such as changes in sectional aerodynamic coefficients influenced by vortex dynamics. In addition, this study focuses on revealing the dominant steady and unsteady effects of the hovering stacked rotor. In order to evaluate the effect of blade-vortex interaction (BVI) on unsteady airloads, a counter-rotating coaxial rotor was used as a comparison rotor. Finally, two-stage neural networks are leveraged to optimize the stacked rotor configuration, and the key insights gained can potentially be used in the future design process of an airfoil or a stacked rotor planform.

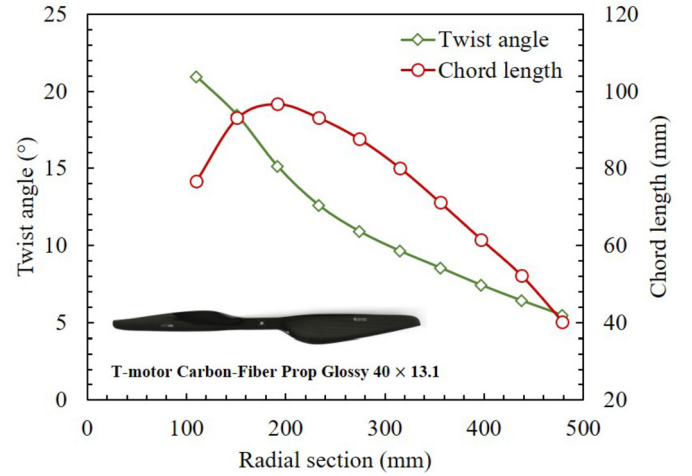


Fig. 1. Detail specifications of baseline rotor (T-motor Carbon-Fiber Prop Glossy 40 × 13.1).

Table 1

Definition and ranges of design variables of stacked rotor configurations.

Variables	Definition	Design range
Z (D)	Stacked distance (unit = diameter (D))	0.1 ~ 0.5 [0.1, 0.15, 0.2, 0.3, 0.4, 0.5]
ϕ (°)	Index angle (positive angle refers to lower blade leading)	-90 ~ 90 [-90, -60, -45, -30, -20, -10, -5, 0, 5, 10, 20, 30, 45, 60, 90]
$\Delta\theta$ (°)	Pitch angle difference	0 ~ 4 [0, 1, 2, 3, 4]

2. Design exploration problem

2.1. Baseline rotor specification

The 'T-motor Carbon-Fiber Prop Glossy 40 × 13.1', which is frequently used in wingless-type UAV and UAM aircraft, was chosen as the baseline rotor. The maximum chord length is 0.1 meters, the tip chord length is 0.04 meters, and the rotor radius is 0.5 meters (20 inches). The chord length and twist angle for each part are shown in Fig. 1. The solidity, σ , of the baseline rotor with two blades is 0.096. The rotor rotates at 3000 rpm, with a tip Mach number of 0.451 and a Reynolds number of 0.988×10^6 .

2.2. Exploration space and performance metric

The index angle (ϕ), the stacked distance (Z), and the pitch angle difference ($\Delta\theta$) between the upper and lower blades were all set as design variables. The design exploration space is depicted in Fig. 2 and Table 1. Firstly, Z refers to the stacked distance between the upper and lower blades. The diameter of the rotor, D, serves as the unit of the stacked distance. The stacked distance has a design range of 0.1 to 0.5. Secondly, the index angle, or ϕ , has a positive value when the lower blade is leading. ϕ has a design range of -90° to 90°. The interval of the index angle was decreased as it approaches 0°. According to previous investigations [11,12], the best aerodynamic performance is shown at the angle of 0°. Therefore, the area around 0° has a narrow range. Lastly, $\Delta\theta$ indicates the difference in pitch angle between the upper and lower blades; in other words, $\Delta\theta = n$ indicates that the pitch angles of upper blades will change by $-n/2^\circ$ and the pitch angles of lower blades by $+n/2^\circ$. The full-factorial approach was used for the design of the experiment (DOE). With the use of a high-fidelity RANS solver, 450 examples were examined.

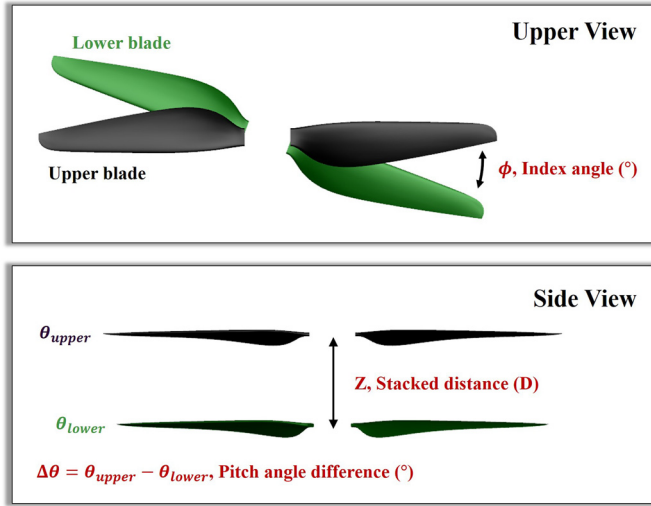


Fig. 2. Design Variables of stacked rotor configurations.

The normalized power loading, PL_{norm} , had been selected as the performance metric of the current design exploration problem, which can be defined as: $PL_{norm} = \text{Normalized Power Loading} = C_T/C_P$ where C_T and C_P represent thrust and power coefficients respectively. For evaluating thrust performance at a constant speed, a normalized power loading was used. Fixed-pitch propellers can be easily trimmed for a particular thrust by altering their rotational speed. Considering that the aerodynamic coefficients do not vary significantly as a function of rotational speed, normalized power loading has been adopted as the comparison metric.

3. Numerical simulation setup

Numerical investigations of aerodynamic performance were performed by using KFLOW [15–17], a Cartesian grid-based three-dimensional, compressible, and viscous solver. With KFLOW, a number of problems involving vortex dynamics had been solved, such as the dominant flowfield of the BVI, propeller wing interaction problems, and delta wing design problems. As for the turbulent closure, the $k-\omega$ shear stress transport model has been used [15,18]. The flow of the stacked rotor in this study was assumed to be fully turbulent. Further details of the numerical schemes are given below.

3.1. Chimera grids system

Chimera grids were used, which consist of a background grid and blade grids. Fig. 3 illustrates the grid system used in this problem. In total, 24 million points were generated for the background grid. In all directions, the far-field boundary exists 25 rotor diameters from the rotation center. The grid spacing near the wake was set at 0.1 mean chord length in order to capture the vortex dynamics with high resolution. Each blade grid consists of 1.8 million points. To simulate the velocity profile within the boundary layer, wall grids were constructed in the condition of $y^+ \leq 1$. 20~25 points exist in the boundary layer, especially 23 points exist at 75% blade span.

3.2. Spatial discretization schemes

To ensure aerodynamic performance reliability and to identify the underlying physics, it is crucial to precisely model the vortex dynamics of coaxial rotorcraft with vortex-dominated characteristics. Given the large computational load of 450 cases, an effective algorithm is necessary. To this end, four high-order accuracy

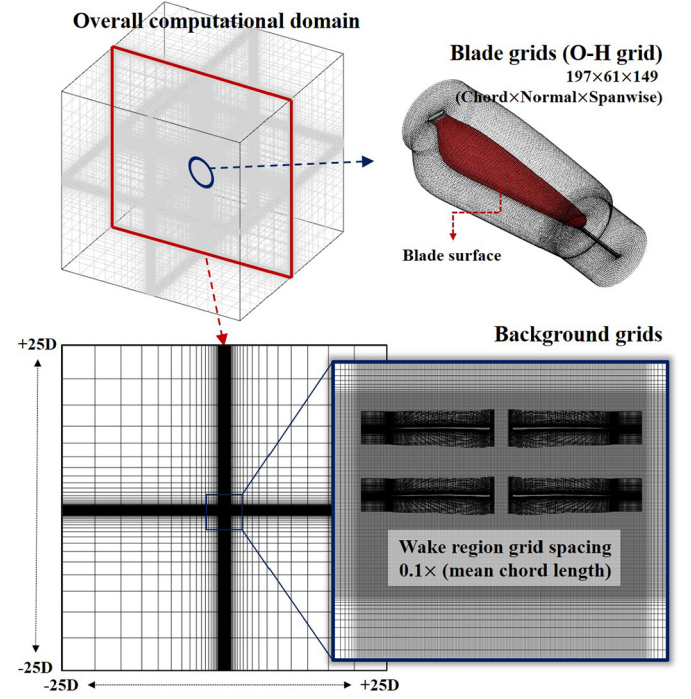


Fig. 3. Chimera grid system of stacked rotor configurations.

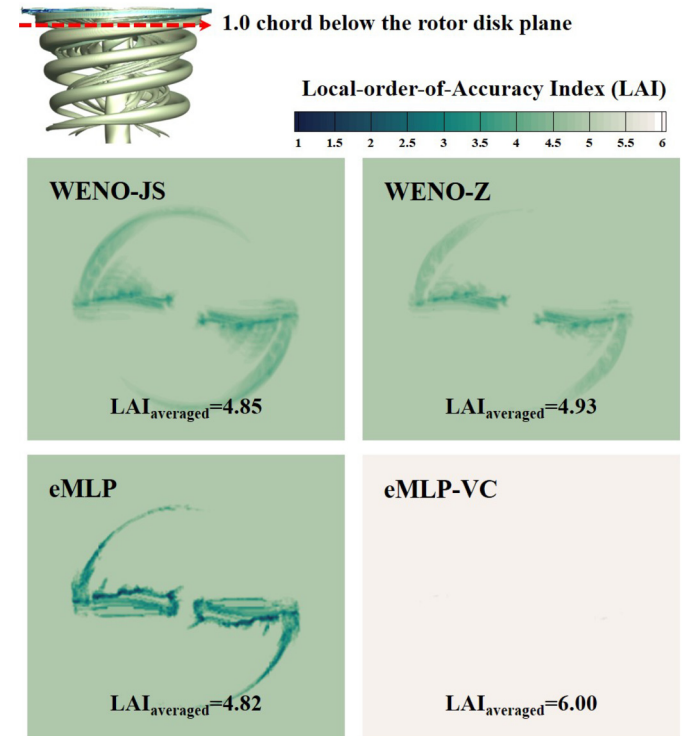


Fig. 4. Local-order-of-accuracy index (LAI) comparison of four different spatial discretization schemes.

schemes with comparable computing power were selected, and a comparison was performed using a baseline rotor. eMLP-type schemes (eMLP [19], eMLP-VC [17]), which are recently developed for the efficient and high-accurate flow analysis, and WENO-type schemes (WENO-JS [20], WENO-Z [21]), which are widely used by rotor aerodynamicists [22,23] for high-accuracy, were selected for the comparison. The results of the comparison are illustrated in Fig. 4.

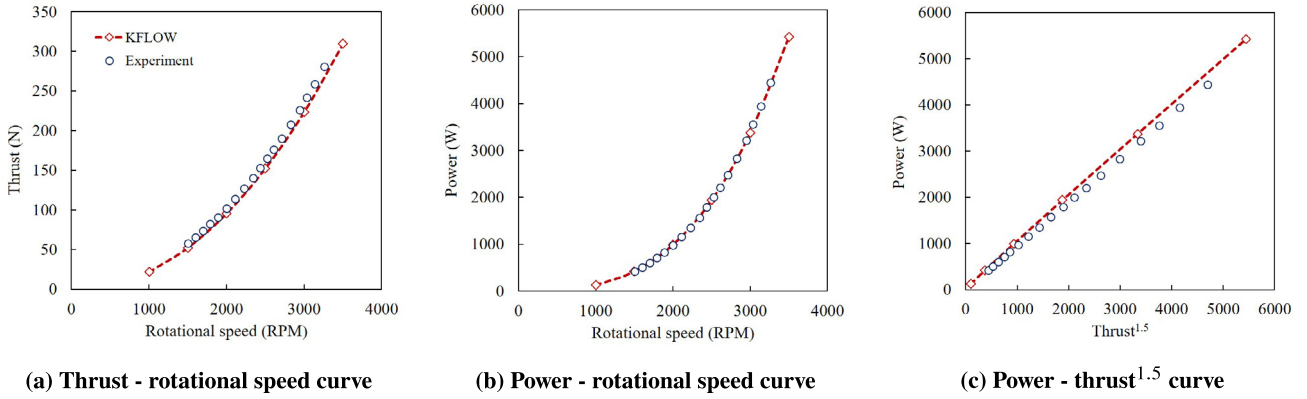


Fig. 5. Comparison of the aerodynamic performance of baseline single rotor with two blades.

The contour of the local-order-of-accuracy index (LAI) at the plane, one chord below the rotor disk, is shown in Fig. 4. The LAI denotes the order-of-accuracy of the numerical scheme in the local discretized domain, and more information is available in Hong et al. [24]. Higher spatial accuracy is correlated with larger LAI values. As observed in Fig. 4, all three of the high-accuracy schemes—aside from eMLP-VC—display accuracy reductions in the vortex trajectory. LAIs of WENO-JS, WENO-Z, and eMLP are 4.85, 4.93, and 4.82, respectively. Considering that they are 5th-order accurate spatial schemes, the LAI should be 5. However, the accuracy is locally decreased by the fact that each scheme detects the vortex gradient as if it was a shock discontinuity and applies artificial dissipation. Because of the decreased accuracy, it is impossible to accurately predict the underlying physics of rotorcraft flowfields. In most cases, the lack of grids causes local spatial accuracy to decrease. However, eMLP-VC exhibits negligible accuracy degradation despite the small number of grids. In addition, due to the inherent low-Mach number adjustment process, eMLP-VC has near sixth-order accuracy in most planes (LAI of eMLP-VC is 6.00). As a result, the eMLP-VC was selected for high accurate aerodynamic analysis of stacked rotor in this research. Additionally, upwind-biased AUSMPW+ was applied as a flux function for the robust calculation [25].

3.3. Temporal integration methods

For 450 DOE cases, steady calculations were conducted. The inversion for flux Jacobian matrix was efficiently performed using the diagonalized alternating-direction implicit algorithm [26]. Courant-Friedrichs-Lewy number was set to 4. In total, 14,000 steps were calculated for the converged solution. Additionally, an unsteady calculation was performed for a particular case with good performance to be compared with that of a counter-rotating coaxial rotor. A dual time-stepping method will be used to reduce the linearization error when performing the unsteady calculation using the 2nd-order accurate backward difference formula. To get the converged solution of the stacked rotor flow, total 20 revolutions were calculated. A first set of 10 revolutions was analyzed at a time step of 2.5° in order to develop the wake and vortex system of the rotor in general. Then an additional set of 10 revolutions was analyzed at a time step of 0.25° . Intel Xeon E5-2690 v4 processors were used for the parallel computation. 48 cores were utilized in one case. The average computation time for steady calculations was 16.7 hours per case. For an unsteady calculation with 0.25° time step, the computation took about 34.4 hours per revolution. The results of experiments conducted by Korea Aerospace Research Institute and Kyungwoon University (not published) were utilized in order to validate the aerodynamic solver. A range of rotational speeds from 1000 revolutions per minute (rpm) to 3500 rpm was used. In Figs. 5a, 5b, thrust, and power curves are shown as a

function of rotational speed. KFLOW solver tends to underestimate thrust to that of the experiments. According to the power-thrust^{1.5} curve (see Fig. 5c), the required power was slightly overestimated. The KFLOW solver shows a slight difference from the experimental value, but the difference is insignificant. Since it is capable of predicting aerodynamic performance trends, it was determined that it is sufficient for the analysis of rotorcraft flowfields.

4. Overall results

KFLOW RANS solver was used to analyze 450 DOE cases. As the configurations with an index angle of 90° are the same as the configurations with -90° , only 420 cases were actually calculated. Furthermore, two single rotors were also calculated for comparison: one with four blades ($\sigma = 0.192$) that has the same solidity as the stacked rotors, and another with two blades ($\sigma = 0.096$) that has the same solidity as the upper and lower blades of the stacked rotors. Both single rotors have the same radius and pitch angles as the stacked rotor configurations with $\Delta\theta = 0^\circ$. The calculations were made based on two assumptions. Initially, it is assumed that the flowfield is steady. The rotor flowfield may be unsteady, however, the unsteadiness has been suppressed for better identification of the underlying physics of aerodynamic performance averaged over time. As the second consideration, the geometry of the blade is assumed to be rigid, and elastic deformation is left out of the calculation. According to previous results on blade elasticity effects in hover conditions [27], the twist angle at the tip changes by approximately 1.3° when blade loading is 0.1, which represents a high thrust condition. The change in twist angle results in thrust reduction. It is important to consider the elastic effect for accurate analysis. However, the aspect ratio of the blade used on the UAM aircraft is much shorter and stiffer than that of a conventional helicopter. Since structural changes in the present baseline blade are expected to be small, elastic deformation was not considered in this study.

The overview of the aerodynamic performances of stacked rotor configurations is presented in Fig. 6. Configurations are arranged in descending order of performance. The normalized power loadings of all stacked rotors are higher than that of a single rotor with four blades of the same solidity. Gains in the performance of stacked rotors compared to that of the four-bladed single rotor range from a minimum of 1.35% to a maximum of 8.65%. However, all stacked rotors are less effective than a single rotor with two blades. Fig. 6 also provides information on the performance of the upper and lower blades of the stacked rotor. Normalized power loadings for all upper blades are greater than those for all lower blades. PL_{norm} on the lower blade has a maximum value of 8.63, which is significantly lower than PL_{norm} on the upper blade (9.00). As compared to single rotors with two blades, the lower blades of all the stacked

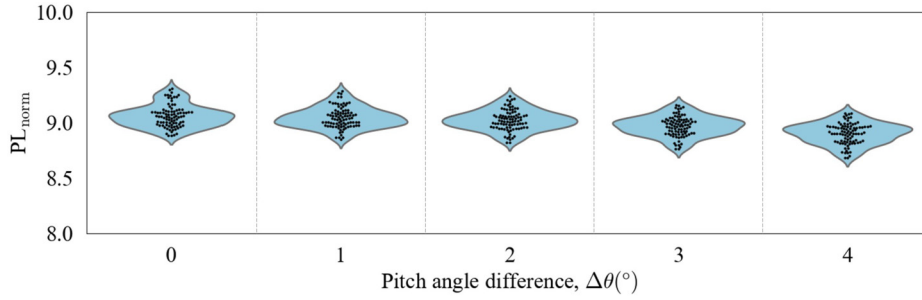


Fig. 7. Violin plot of normalized power loading along the pitch angle difference, $\Delta\theta$.

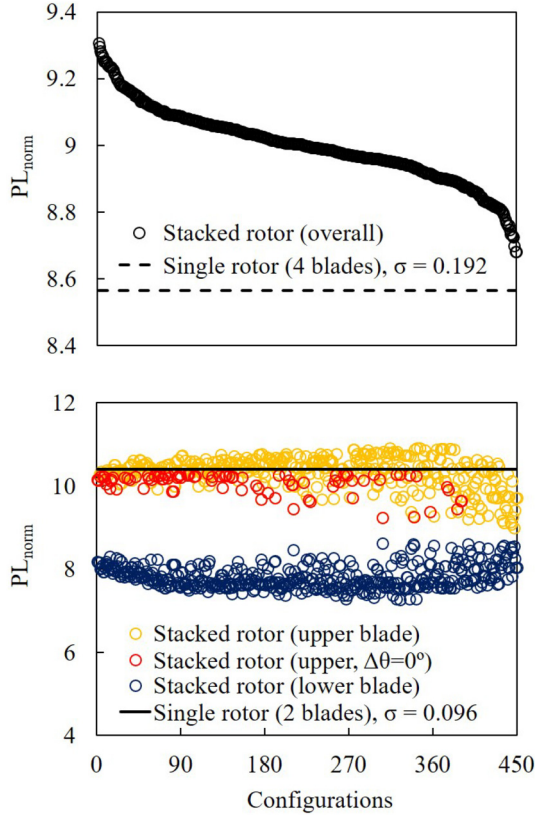


Fig. 6. Normalized power loading of whole stacked rotor configurations.

rotors exhibit diminished aerodynamic performance. For the upper blades, configurations with the same pitch angle as the single rotor have lower aerodynamic performance. These results are consistent with those of previous studies [10,12], and it is apparent that each blade has a substantial influence on the others.

Normalized power loadings of every DOE case were plotted in Fig. 7 as a function of pitch angle differences. The violin plot, which is the type of density curve, shows the full distribution of power loading. The black dots inside the violin plot represent PL_{norm} of each case. A region with more data points will have a wider area in the plot. As the violin plot presents the density curve on the horizontal axis, it means that the standard deviation is small when horizontally elongated and large when vertically elongated. As shown in Fig. 7, PL_{norm} decreases slightly as the pitch angle difference increases. It should be noted, however, that similar distribution patterns are shown in all pitch angle differences. In other words, the degree of influence on the PL_{norm} is the same for all pitch angle differences. Though the pitch angle difference is a variable that significantly affects the inflow strength of each blade and the strength of the tip vortex, it has little impact on the phys-

ical phenomena that will be discussed in more detail in the next chapter.

5. Underlying physical phenomena

Rotorcraft with a single rotor pose complex and nonlinear aerodynamic interactions around the rotor disk. As the vortex generated by the rotor blade is very close to the airframe, it has a substantial effect on the overall aerodynamic performance of the helicopter. This vortex reduces the effective angle of attack of the rotor blade and degrades aerodynamic efficiency. Moreover, various aerodynamic interactions, such as vortex-vortex, vortex-blade, and vortex-miscellaneous surface interactions, substantially impact the aeroacoustic noise and vibration of the rotorcraft. However, the flow physics of a stacked rotor configuration is more complicated than that of a single rotor, as the two main propulsors are in close proximity and on the same axis. The vortices of the upper and lower blades have a strong aerodynamic influence on each other, leading to a highly nonlinear and intricate flowfield. Two crucial effects, inflow and BVI in lower blades, significantly impact the aerodynamic performance of stacked rotors. These effects must be studied to better understand the aerodynamic behavior of stacked rotors and to design more efficient and less noisy rotorcraft.

5.1. Inflow effects

The inflow is defined as flow passing through the surface of a rotor, primarily caused by pressure differences between its upper and lower surfaces. The inflow is strongly influenced by the downwash generated by the vortex generated at the tip of the rotor blade. Coaxial rotors have both upper and lower blades, which can also affect each other's inflow. According to the study by Uehara et al. [10], stacked rotor performance is significantly affected by the inflow induced on one rotor by another rotor. When considering coaxial rotors, it is necessary to understand the mechanism of inflow generation of both upper and lower blades.

In the case of the upper blade, as shown in Fig. 8a, a suction effect can be observed in which the lower blade absorbs the wake of the upper blade. This effect makes the inflow of the upper blade stronger. Therefore, the effective angle of attack is decreased (see Fig. 8c), and the aerodynamic performance has been significantly reduced when compared to the aerodynamic performance of a single rotor with the same pitch angle.

Fig. 9 presents flowfields for a single rotor with two blades and a stacked rotor configuration with $(Z, \phi, \Delta\theta) = (0.1D, 90^\circ, 0^\circ)$. An increase in inflow can clearly be seen in the vertical miss distance, which is the vertical distance between the blade and the tip vortex of the previous blade. The single rotor with two blades has a miss distance of $0.05D$. Conversely, the upper blade of a stacked rotor has a miss distance of $0.1D$, which is twice the miss distance for a single rotor. Due to the lower blade of the stacked rotor, the inflow

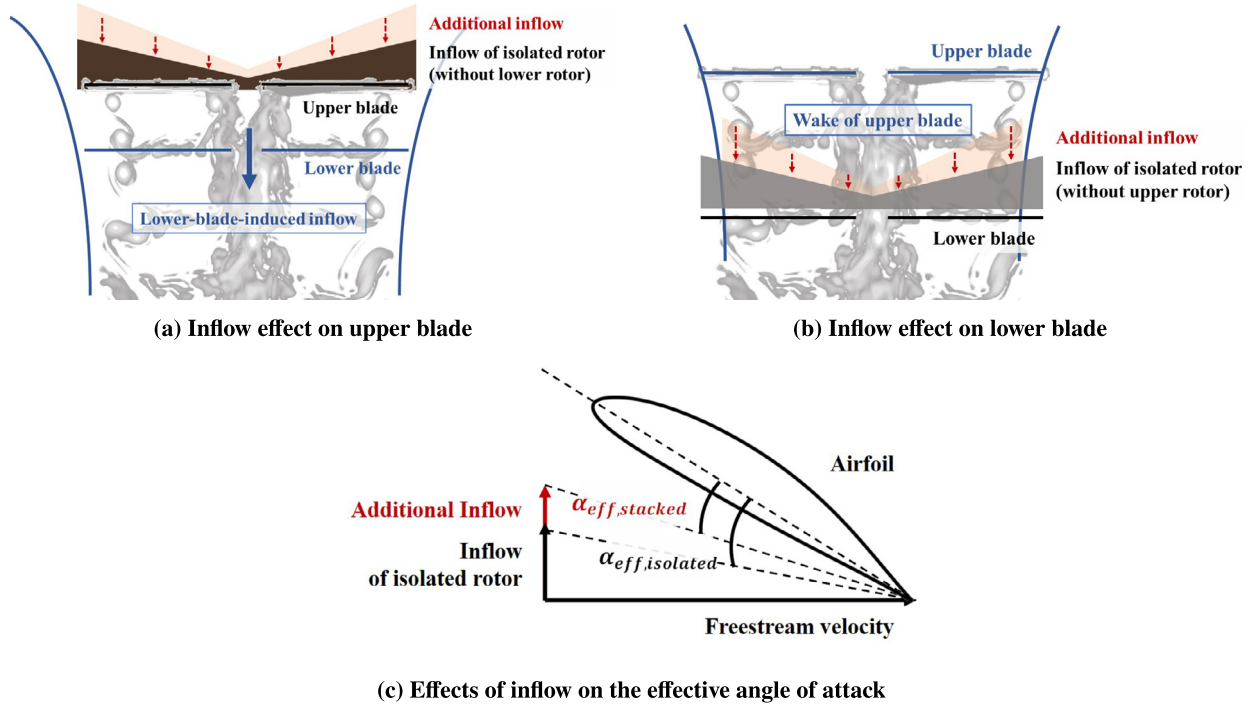


Fig. 8. Schematics of inflow effect on upper and lower blades.

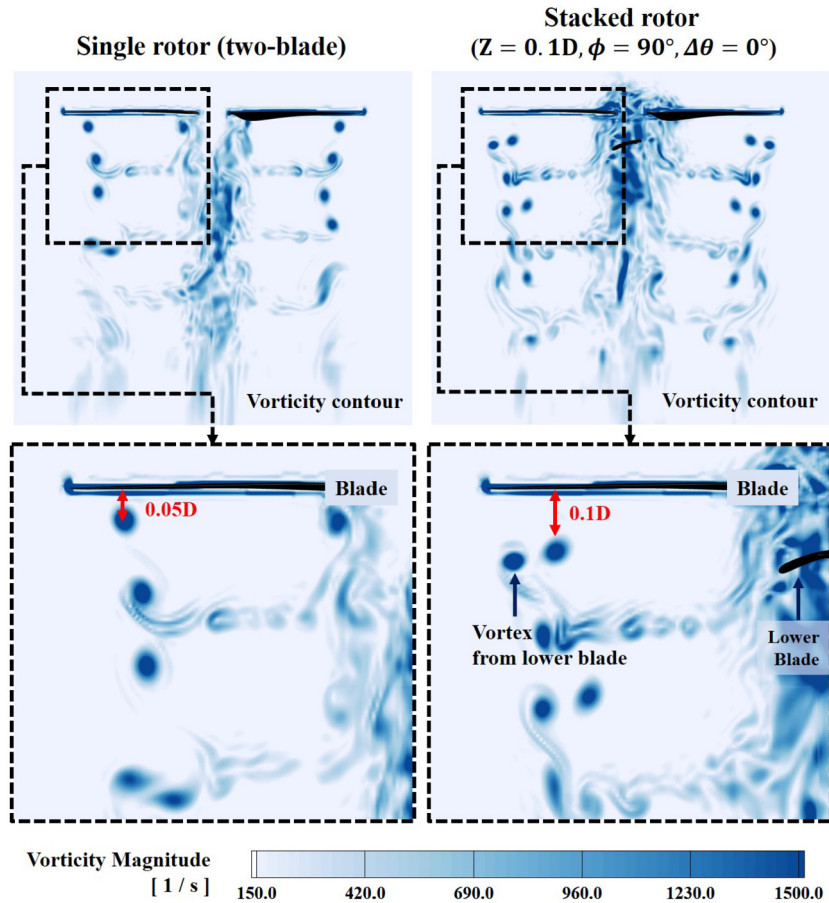
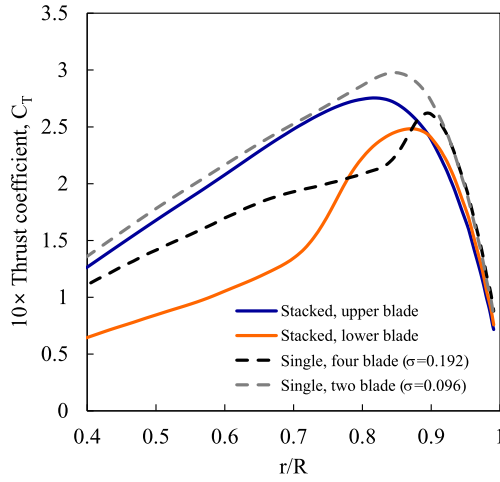


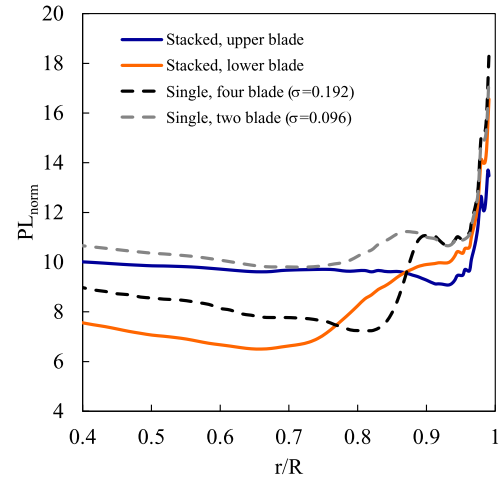
Fig. 9. Vorticity contours of single rotor and stacked rotor.

to the upper blade becomes more intense, resulting in an increased advection velocity of the vortex and thus a longer miss distance to the upper blades.

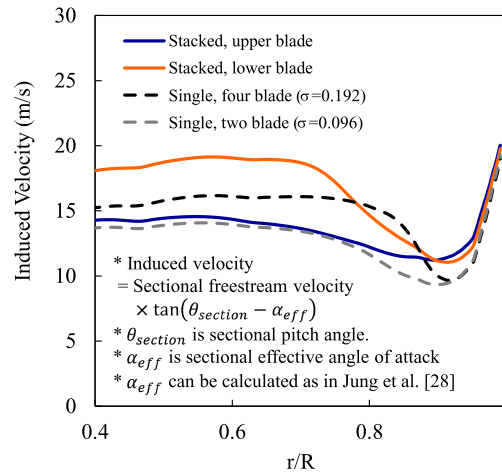
For all stacked rotor configurations with $\Delta\theta = 0^\circ$, Fig. 10 shows the miss distance of upper blades as a function of the stacked distance. All configurations provide miss distance rang-



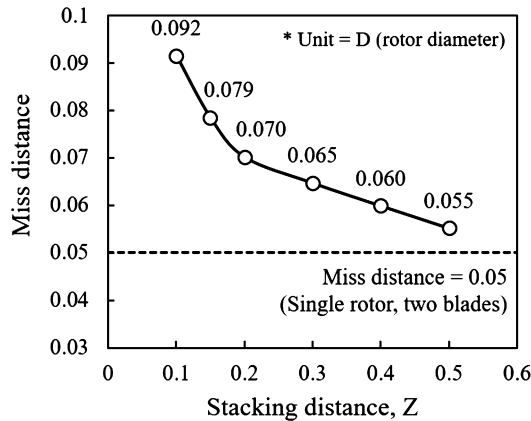
(a) Thrust coefficients comparison



(b) Normalized power loading comparison



(c) Induced velocity comparison

Fig. 11. Aerodynamic loads distribution of upper and lower blades, $(Z, \phi, \Delta\theta) = (0.1D, -90 \sim 90^\circ \text{ averaged}, 0^\circ)$.Fig. 10. Miss distances of upper blades ($\Delta\theta = 0^\circ$).

ing 0.055D~0.092D for the upper blades, indicating that the lower blade impacts the inflow of the upper blade. A decrease in miss distance is observed as the distance between the upper and lower blades increases. Therefore, as long as the physical distance between the upper and lower blades is sufficiently long, the upper blade is less likely to be affected by the lower blade.

Inflow is more apparent on the lower blade than on the upper blade. The wake of the upper blade merges directly with the inflow of the lower blade (see Fig. 8b). Stronger inflow reduces the aerodynamic performance of the lower blade by reducing the effective angle of attack almost everywhere except near the tip ($r/R \geq 0.9$). Fig. 11 provides an overview of the distribution of thrust, normalized power loading, and induced velocity within stacked rotor configurations $(Z, \phi, \Delta\theta) = (0.1D, -90 \sim 90^\circ \text{ averaged}, 0^\circ)$. The effective angle of attack was calculated by using the method used in Jung et al. [28], and the induced velocity in each section was determined. The sectional pitch angle and the freestream velocity were also considered in the calculation of induced velocity. By averaging all configurations with various index angles, the influence of the index angle has been eliminated. The information on single rotors has also been added for comparison purposes. As the four-blade single rotor has a much closer miss distance than a two-blade single rotor, the effect of the tip vortex is noticeable in the region of $r/R \leq 0.9$. Similarly, in the $r/R \leq 0.9$ region, the thrust and the power loading of the lower blade were smaller than those of the upper blade. The decreases are primarily due to the upper blade wake, as shown in Fig. 8b. As shown in Fig. 11c, the induced velocity of the lower blade increases in the inner region ($r/R \leq 0.9$). However, as the wake of the upper blade is relatively concen-

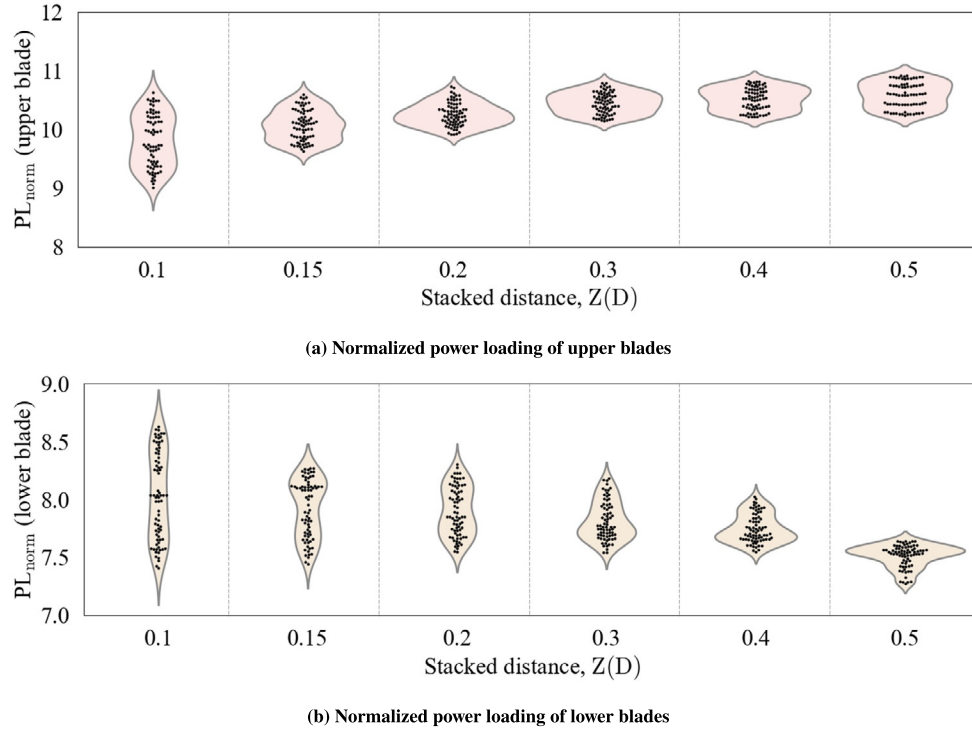


Fig. 12. Violin plot of normalized power loading along the stacked distance.

trated in the inner portion of the blade, it turns out to have little impact on the miss distance of the lower blade.

Stacked rotors exhibit different aerodynamic performances depending on the stack distance, as shown in Fig. 12. Violin plots in Fig. 12 explain the effect of stacked distance on each blade. Different patterns of data distribution are observed at each stacked distance. The standard deviation is large in both the upper and lower blades at $Z = 0.1D$, but it decreases with increasing stacked distance; When the stacked distance is small, the index angle and pitch angle difference have a more substantial impact than the stacked distance itself. As the stacked distance increases, however, the effect of the index angle and pitch angle difference diminishes, and the effect of the stacked distance becomes more significant. Average PL_{norm} on the upper blade increases as Z increases. The performance change is less pronounced as the stacked distance approaches $0.3D$, which indicates that there is no longer a change in the effect of the lower blade on the inflow of the upper blade inflow. In contrast, the PL_{norm} on the lower blade decreases as Z increases. As a consequence of the continuous development of the wake of the upper blade, a stronger inflow occurs, causing this decrease in power loading on the lower blade.

5.2. Blade-vortex interaction effects

Another dominant effect on stacked rotor power loading is BVI, which is the interaction between the tip vortex of the upper blade and the lower blade. In conventional rotor blades, BVI is one of the main sources of aeroacoustic noise and vibration, and should therefore be avoided; however, it is different in stacked rotors. BVI occurs periodically in conventional helicopters, but continuously in stacked rotor configurations. The tip vortex of the upper blade is adjacent to the lower blade in the stacked rotor configuration, producing additional upwash and downwash. Fig. 13 illustrates downwash occurring in the inner region of the lower blade and upwash appearing in the outer region. Local reductions in the effective angle of attack occur in the downwash areas, while the opposite is

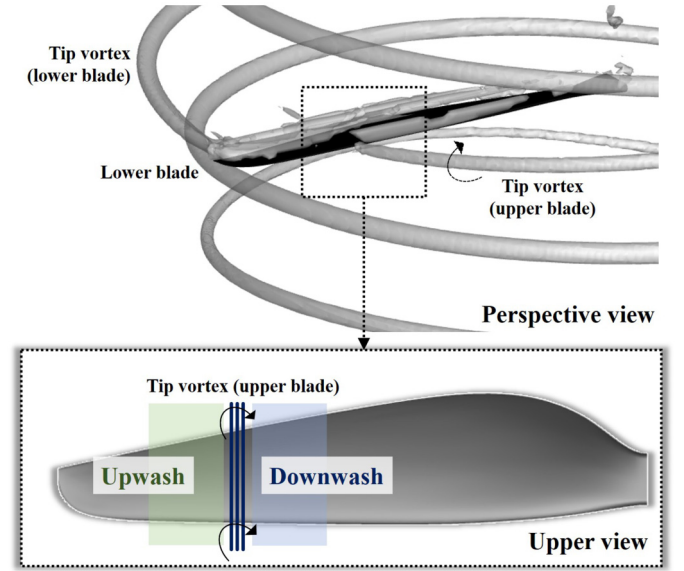


Fig. 13. A schematic of BVI effect on lower blades.

observed in the upwash areas. In consequence, the higher the local effective angle of attack, the higher the power loading. Upwash and downwash caused by the tip vortex affect aerodynamic performance in opposite directions; however, the upwash region exhibits faster sectional free stream velocity than the downwash region, resulting in an advantage in overall aerodynamic performance. In other words, the vortex interacting with the lower blade of the stacked rotor plays an important role as the source of lift of the aircraft. It should be noted, however, that if the stall margin is insufficient, there is a risk that the blade will fall into the stall region due to BVI.

In Fig. 14, the PL_{norm} is presented along the index angle for all DOE cases. In stacked rotors of each Z and $\Delta\theta$, the maximum

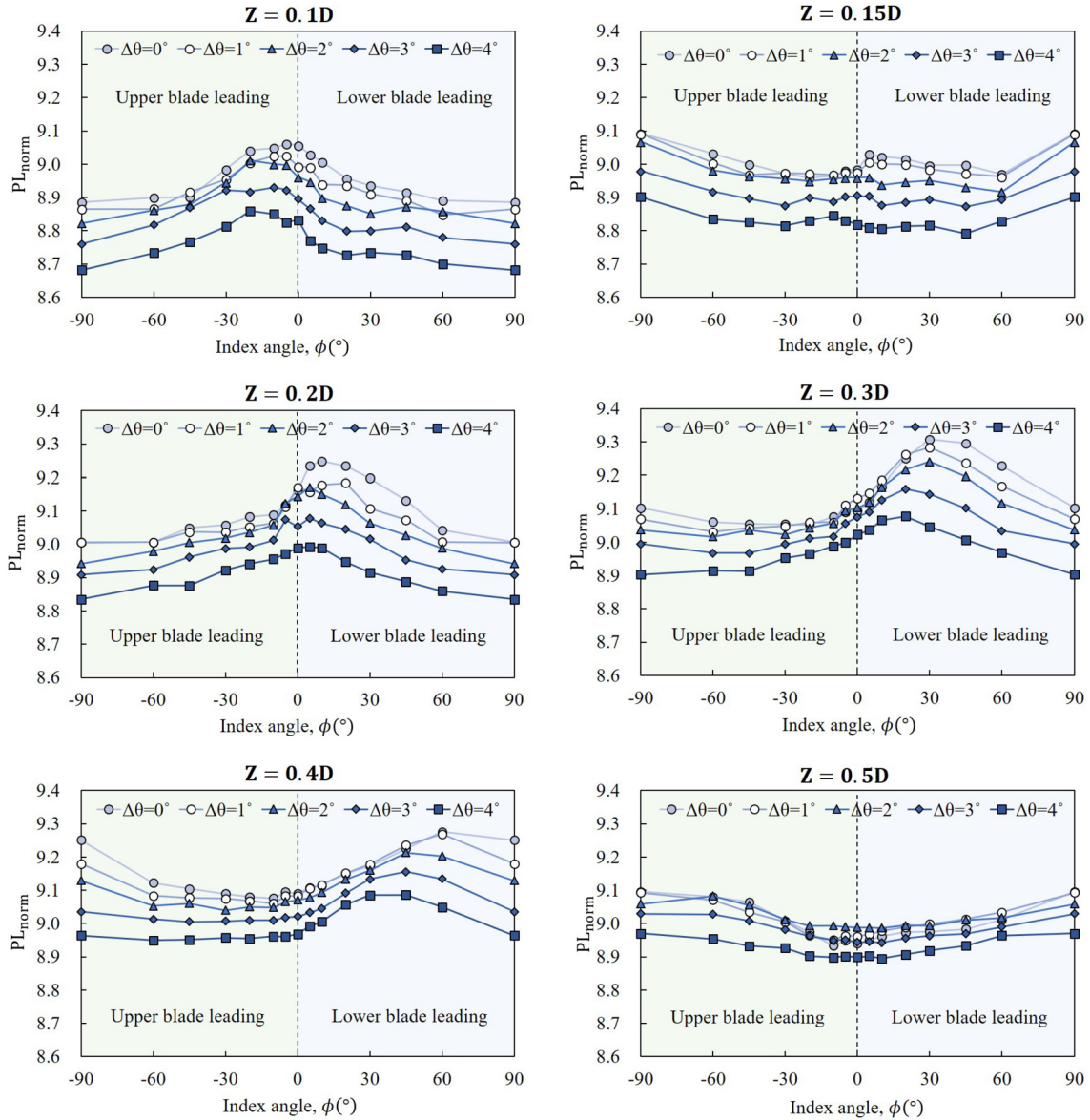


Fig. 14. Normalized power loading comparison of all DOE cases along the index angle, ϕ .

power loading occurs at a particular index angle. For $Z = 0.1D$, the maximum power loading will be achieved when the index angle is between -5° and -20° , where the upper blade leads. For $Z = 0.15D$, the power loading is maximized at $\Delta\theta = 90^\circ$. The index angle with the maximum PL_{norm} gradually moves towards the lower blade leading area as the stacked distance increases. The power loading is maximized at $5^\circ \sim 10^\circ$ at $Z = 0.2D$, $20^\circ \sim 30^\circ$ at $0.3D$, and $30^\circ \sim 90^\circ$ at $0.4D \sim 0.5D$.

Changing the index angle can lead to two notable effects. In the study by George Jacobellis et al. [12], an airfoil interaction effect is observed when the upper and lower blades are sufficiently close together. The close proximity of the two airfoils results in a mutual influence on their bound vorticity, thereby affecting the effective angle of attack and, therefore, aerodynamic performance. Secondly, there is the BVI effect, which occurs when the lower blade interacts with the tip vortex of the upper blade. This phenomenon is illustrated in Fig. 13. In general, these two effects are independent of each other, and the smaller the gap between the rotors, the greater the influence of the airfoil interaction. However, within the design space considered in this study, the BVI effect appears to be more dominant (see Figs. 15, 16).

Fig. 15 depicts the flowfields of stacked rotors, which have the best aerodynamic performance within each stacked distance. In order to visualize the vortices, the iso-surface method was used with the Q-criterion value, $10 \times U_{\text{tip}}^2$. U_{tip} is the speed of blade tip section. Each of the cases shown in Fig. 15 involves BVI in the lower blade. A clear illustration of the BVI effect can be found in Fig. 16, which illustrates the distribution of aerodynamic coefficients of the stacked rotor configurations $(Z, \phi, \Delta\theta) = (0.3D, -90^\circ \sim 90^\circ, 2^\circ)$. The blue lines and shaded areas represent thrust coefficients, while the green lines and shaded areas represent normalized power loading. A shaded area indicates the aerodynamic loads of all cases with $\phi = -90^\circ \sim 90^\circ$. The $\phi = 30^\circ$ and $\phi = -30^\circ$ cases are representative cases that show the BVI effect prominently and are represented by solid and dotted lines, respectively. Please note that each case does not generate the same thrust because it is not a trimmed solution. $\phi = 30^\circ$ case exhibits the most notable BVI effect, as well as the highest power loading. $\phi = -30^\circ$ case, however, does not exhibit BVI and has the smallest power loading. Differences in solid lines and dotted lines show the BVI effect on aerodynamics. When BVI occurred, in the outer region ($r/R \geq 0.7$), power loading improved substantially, whereas in the inner region ($r/R \leq 0.7$), power load-

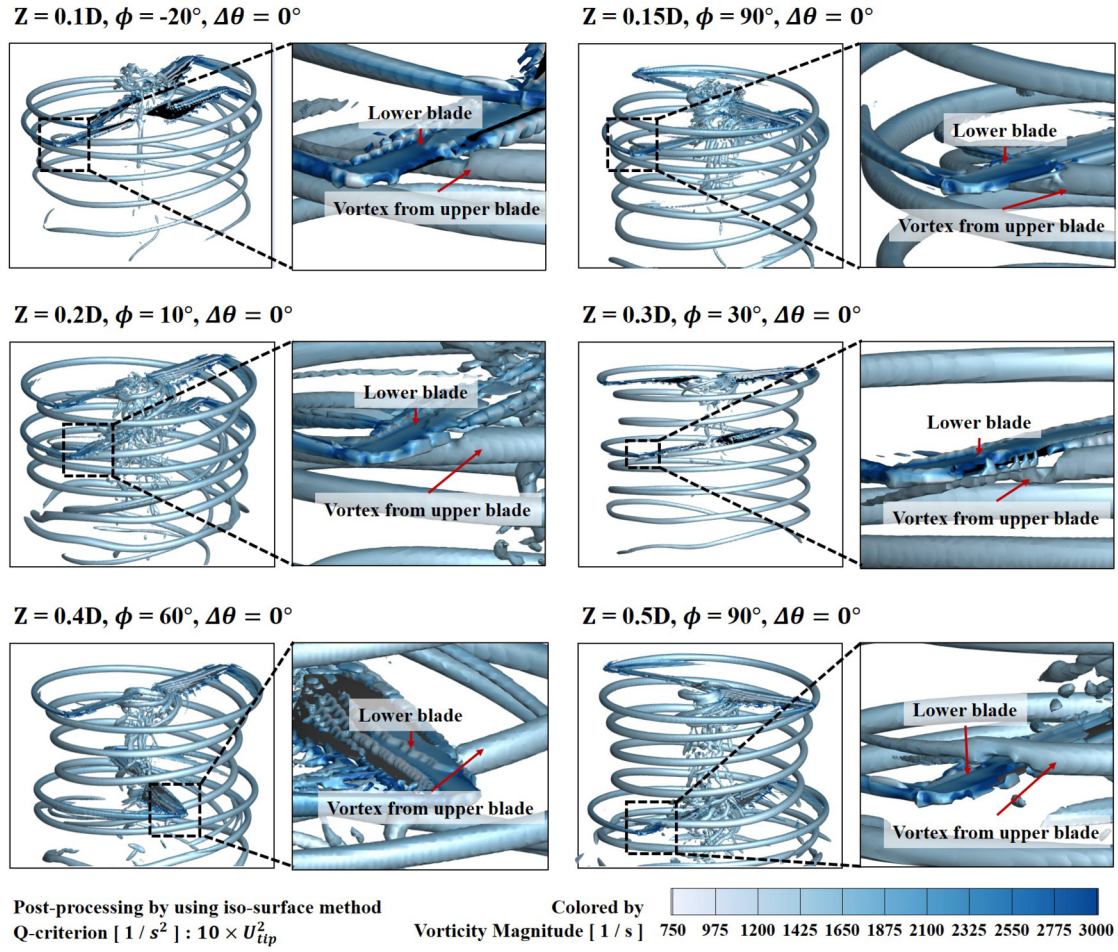


Fig. 15. Flowfields of best configurations in each stacked distance (All configurations have BVI in the lower blades).

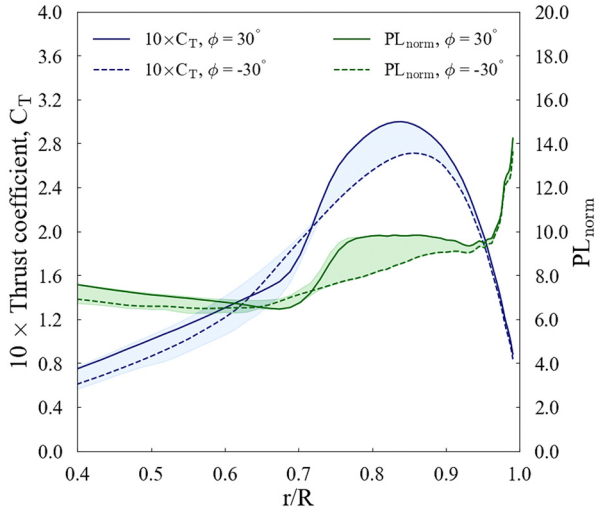


Fig. 16. Aerodynamic coefficients distribution of lower blades (parameters of stacked rotor configurations are $(Z, \phi, \Delta\theta) = (0.3D, -90^\circ \sim 90^\circ, 2^\circ)$). (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

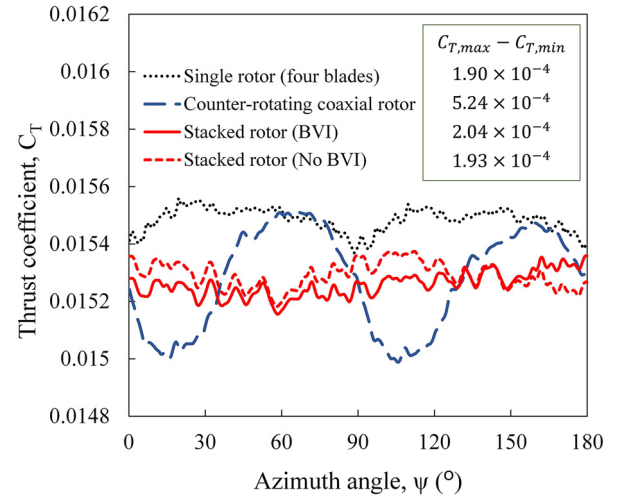


Fig. 17. Thrust history of four different rotors.

5.3. Unsteady effect

ing decreased. The upwash created by the vortex increases the effective angle of attack, which increases the power loading. As increased PL_{norm} in the fast outer region has a more significant effect on improving the entire performance of the aircraft, these results support the hypothesis that the BVI effect considerably influences the stacked rotor.

The best design point that produces maximum power loading is the design variable set at $(Z, \phi, \Delta\theta) = (0.3D, 30^\circ, 0^\circ)$. A stacked distance of $0.3D$ allows the inflow suction effects on the upper blade and lower blade to be optimally adjusted. Increased miss distance of the upper blade reduces the degradation of performance. Furthermore, the best design point has BVI in the lower blade. The BVI plays an essential role in improving aerodynamic performance

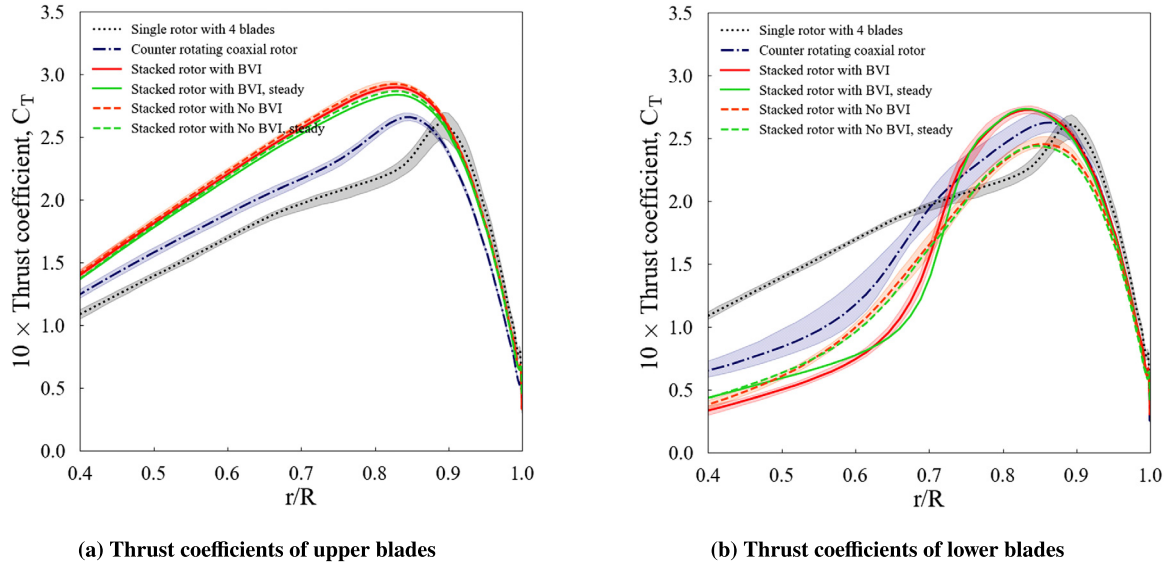


Fig. 18. Comparison of sectional thrust coefficients for four different rotors.

from a steady-state perspective. The upwash induced by BVI locally increases the effective angle of attack, which resultantly increases the power loading on the lower blade. However, the BVI may have secondary effects, such as causing undesirable aeroacoustic noise and vibration. Further calculations were conducted to evaluate the effects of BVI using an unsteady RANS solver in addition to the steady-state calculation. The steady solver used in previous DOE analyses did not take into account the changes in the flowfield over time. A limitation of the steady calculation is the inability to describe the random behavior of vortices. As a result of this inability, the vortex trajectory can differ between the steady and unsteady results. In the case of the best design point, due to differences in vortex trajectory predictions, the index angle at which BVI occurs varies by approximately $\phi = 15^\circ$. When considering the differences between two solvers and temporal discretization errors, the differences are quite reasonable. These deviations are inevitable in experiments as well.

For an accurate prediction of the BVI effect, the configuration with the index angle offset by 15° , $(Z, \phi, \Delta\theta) = (0.3D, 45^\circ, 0^\circ)$, was calculated through URANS solver. As a comparative example, an additional stacked rotor configuration was also examined, where the stack spacing and pitch angle differences are identical to the best case, with the index angle changed $(Z, \phi, \Delta\theta) = (0.3D, 0^\circ, 0^\circ)$. No BVI is present in this configuration. Furthermore, a single four-bladed rotor which has the same solidity as stacked rotors, and a counter-rotating coaxial rotor with a stacked distance of $0.3D$ were examined. By adjusting the pitch angle of the upper and lower blades, thrust and power can be adjusted for the counter-rotating coaxial rotor. The target thrust was determined to be the thrust of the best DOE case. Convergence was achieved when the thrust and power values were both within 1%.

Fig. 17 illustrates the thrust histories of four different rotors. As a result of the intersection of the lower and upper blades, the counter-rotating coaxial rotor exhibits a marked thrust fluctuation at a frequency of $4/\text{rev}$. On the other hand, in the stacked rotor, the degree of fluctuation at $4/\text{rev}$ is comparable to that at high frequencies. In particular, the stacked rotor with BVI produces a comparable degree of fluctuation to the rotor without BVI, even though BVI remains in the lower blade of the rotor. It is also noteworthy that the difference between the maximum and minimum thrust values is approximately three times greater for the counter-rotating coaxial rotors than it is for the stacked rotors. These re-

sults suggest that the stacked rotors would generate a lower level of tonal noise than the counter-rotating rotors.

Fig. 18 shows the sectional thrust coefficients for four different rotors. Dark lines indicate the mean thrust coefficients and the interval between minimum and maximum values is displayed in light colors. For the stacked rotor configurations, steady results are also presented. The upper blade showed little difference between steady and unsteady data, while the lower blade showed a slight difference in the inboard section. It is due to the fact that the wake of the upper blade appears relatively unsteady in the inboard section. Steady solvers are unable to capture the effects of unsteady wakes, resulting in differences in load. It should be noted, however, that these differences had little effect on the overall aerodynamic performance. Sectional loading of the upper blade is not significantly affected by BVI as shown in 18a. All coaxial rotors exhibit small thrust coefficient fluctuation in the upper blades. Meanwhile, BVI clearly affects the lower blades. A tip vortex created by the upper blade increases the outer region thrust of the lower blade and decreases the inner region thrust. It follows the same pattern as Fig. 16. One thing to note is that regardless of its presence, BVI in stacked rotors has little impact on the difference in min-max thrust values. There is a similar amount of thrust fluctuation in the lower blade of a stacked rotor with and without BVI. Meanwhile, a counter-rotating coaxial rotor, in which temporary BVI occurs four times per revolution, exhibits a different aspect. Compared to the other rotors, the lower blade exhibits relatively large fluctuations in airloads. A clearer illustration of this effect can be found in Figs. 19, 20.

Figs. 19, 20 describe the thrust coefficient and the derivative of the thrust coefficient for each azimuth angle (ψ). According to Fig. 19, stacked rotors and the single rotor exhibit similar thrust distributions with virtually no variance in thrust. Neither the upper nor lower blades show any fluctuations in thrust. On the other hand, in the case of a counter-rotating coaxial rotor, the thrust changes every 90° when the blades cross at every 90° . Also, the tip vortex of the upper blade interferes with the lower blade, leading to a change in thrust. The thrust history in Fig. 17 confirms this result.

The measurements of thrust fluctuation ($dC_T/d\psi$) are expressed in Fig. 20. The single rotor showed the largest averaged thrust fluctuation.

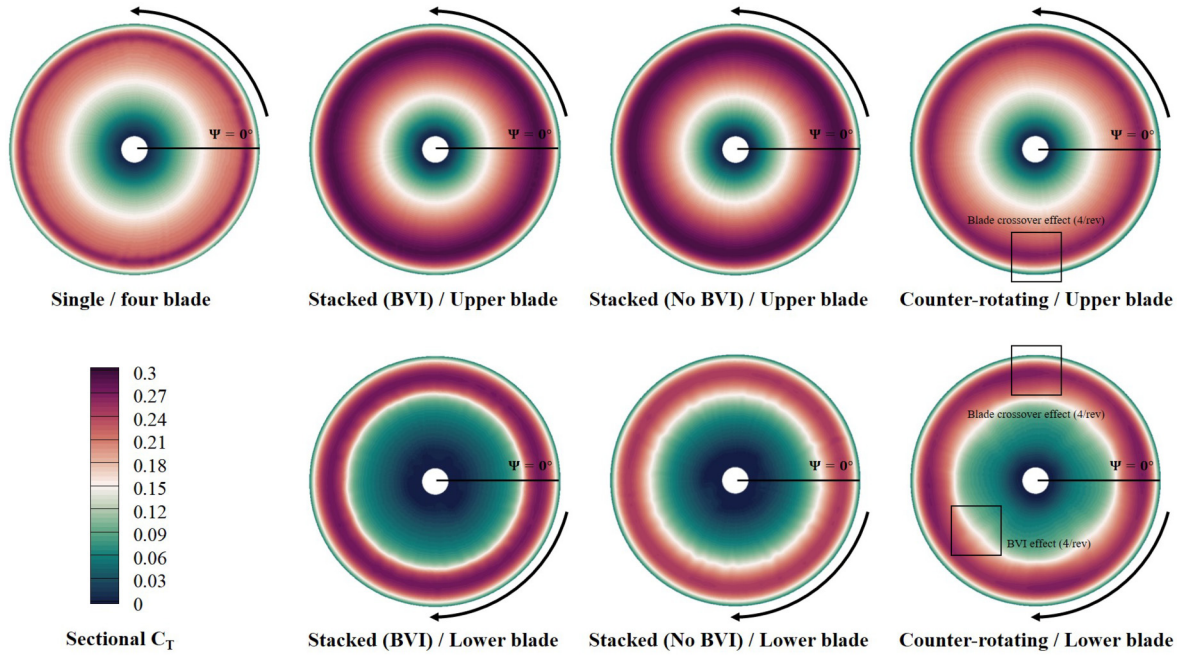


Fig. 19. Thrust coefficients distributions along the azimuth angle (ψ) of four different rotors.

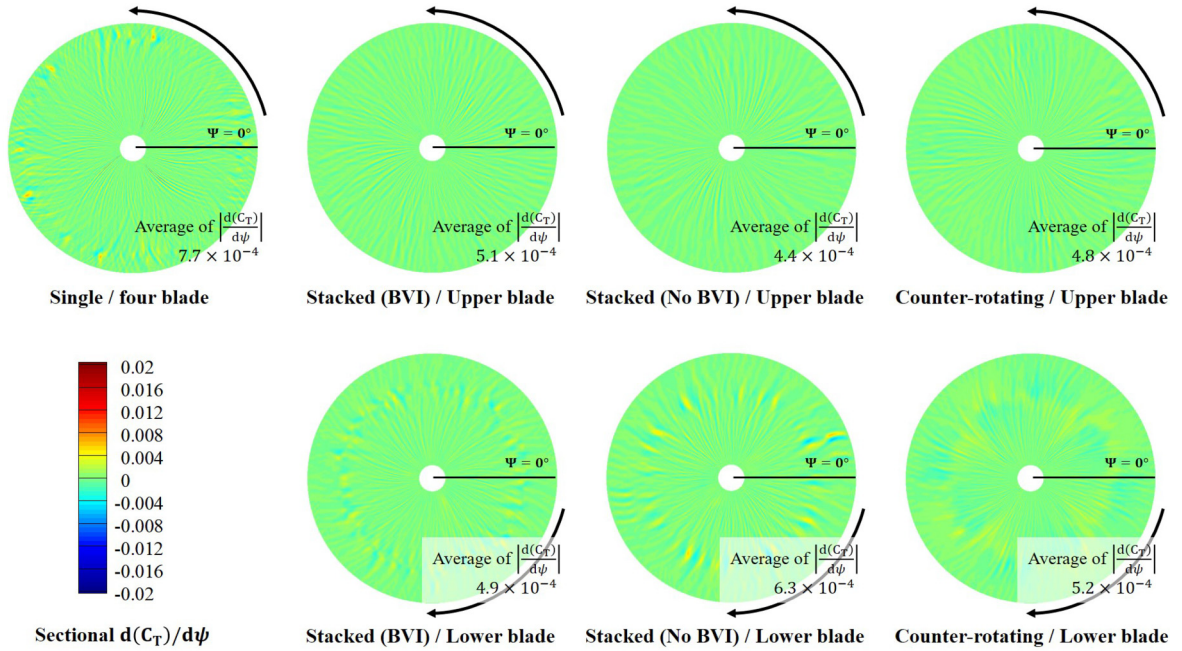


Fig. 20. Derivative of thrust coefficients ($dC_T/d\psi$) along the azimuth angle (ψ) of four different rotors.

tuation ($dC_T/d\psi = 7.7 \times 10^{-4}$) among the four different rotors. Due to the proximity of the vortex produced by the preceding blade to the following blade, a larger amount of thrust perturbation is generated to the single rotor. On the other hand, coaxial rotors with co-rotating and counter-rotating blades have half the number of blades in one plane compared to the single rotor. Consequently, the vortex of the prior blade has a much smaller impact on the following blade. Moreover, since the upper and lower blades are separated by a distance of more than 0.3 diameter, the vortex produced by the upper blade becomes weaker. The weakening of vortex interaction results in small perturbation of thrust. Therefore, the thrust fluctuation value of the coaxial rotor has been reduced.

BVI has negligible effects on thrust fluctuation in stacked rotor configurations, as shown in Fig. 20. As the relative position of the vortex does not alter with time when BVI continues in the lower blade, the perturbation remains relatively unchanged. The BVI generated in a stacked rotor will not generate any substantial thrust fluctuations as long as the flow conditions do not change, such as during a strong wind gust. Thus, the coaxial rotor may produce less high-frequency noise such as broadband noise in comparison with a single rotor. There will, however, be increased thrust fluctuation if the stacked distance is less than 0.3D, since the vortex generated by the upper blade will be less dissipated. A further study will be required to evaluate the relevance of stacked distances.

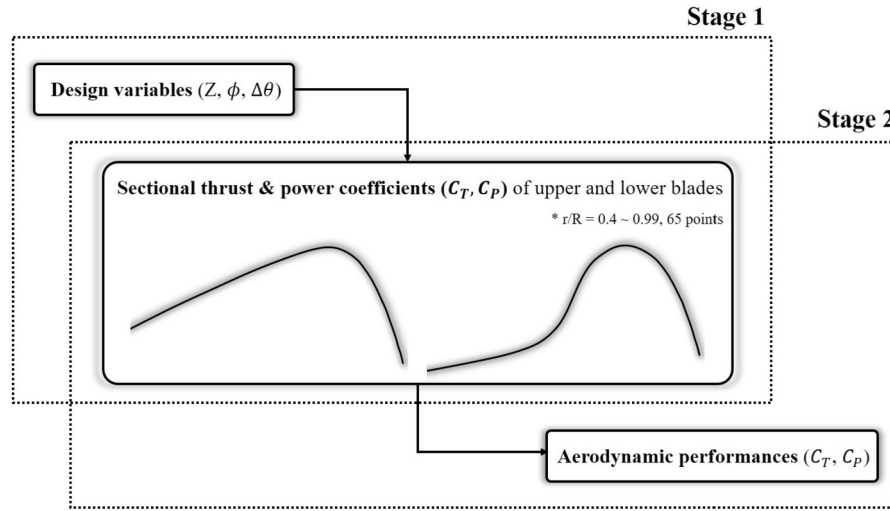


Fig. 21. Flowchart of the two-stage neural networks.

Table 2
Design parameters and PL_{norm} of best stacked rotor configurations.

	Z (D)	ϕ (°)	$\Delta\theta$ (°)	C_T	C_P	PL_{norm}
Single (4 blade)	-	90.0	-	0.01573	0.001836	8.57
Stacked (Best DOE case)	0.30	30.0	0.0	0.01527	0.001641	9.31
Stacked (Optimized)	0.36	50.1	0.0	0.01514	0.001616	9.37

6. Design optimization

As a final step, design optimizations were performed for aerodynamic performances. A surrogate model of neural networks was employed to facilitate an efficient optimization process, reducing the computational cost associated with the high-fidelity CFD solver [29], resulting in near-real-time prediction of the sectional aerodynamic coefficients of various stacked rotor configurations. An optimization of the design was conducted using a single objective function, which was determined as follows:

$$\text{Maximize } PL_{\text{norm}} = C_T/C_P \quad (1)$$

6.1. Methodology

The flowchart in Fig. 21 depicts how the two-stage neural networks are implemented in the present design optimization procedure. In the first stage, the three design variables were set as the input and the sectional thrust and power coefficients were set as the output. Due to strong nonlinearity and low freestream velocity, the regions with r/R 0.4 were excluded. Second-stage neural networks were constructed using the outputs of the first-stage neural network as inputs. As a result, the overall thrust coefficient and power coefficient of the upper and lower blades were obtained. Leveraged neural networks consist of two hidden layers with 50 nodes, LeakyReLU [30] as the activation function, and Adam [31] as the optimization approach. A learning rate scheduler with an initial learning rate of 0.001 and a multiplicative factor of 0.7 per 1000 epochs is also employed [29]. In total, 30,000 epochs were performed.

A dataset of 400 cases was used for training the neural networks above. The validation of the trained neural networks was performed on 50 cases, and some of the results are shown in Fig. 22. The circles represent the predictions from the two-stage framework, while the lines represent the results of CFD. All test datasets produced superior results, as shown in Fig. 22. Both neural networks show coefficients of determination (R^2) of 0.991 and 0.987, respectively.

6.2. Optimized designs and design rule extraction

A summary of the optimization results can be found in Table 2. An optimization of Eq. (1) was conducted based on the trained networks. Through optimization, the best case was obtained with a normalized power loading of 9.37, which resulted in an improvement of 9.33% in efficiency over a single rotor. Table 2 shows the combination of design variables. The stacked distance and index angle of the optimized case are larger than that of the best DOE case, while the pitch angle differences of the two cases are the same. Increases in stacked distance and index angle indicate that the inflow and BVI effects are being optimized, as shown in Fig. 23 as well.

Fig. 23 compares the sectional distributions of the best DOE and optimized cases. There is no noticeable difference between the best DOE case and the optimized case in the upper blades, but there is a significant difference in the lower blades. The optimal design produces more thrust on the outboard side of the lower blade and less thrust on the inboard side. The difference in power coefficient, however, is almost negligible. Therefore, the optimal design possesses a higher thrust coefficient on the faster outboard side, which has a positive effect on aerodynamic efficiency.

So far, the single-objective design optimization was carried out using two-stage neural networks. Taking a step further, useful and practical design rules can also be extracted using these trained regression models [32]. In contrast to counter-rotating coaxial rotors, stacked rotors offer the advantage of changing their geometry in accordance with flight conditions without considering torque balance. As a result, for an individual flight profile, the rotor geometry that is most efficient under that particular condition can be used without considering any trade-offs between other profiles. By applying the current two-stage prediction framework to the multi-objective design problem of maximizing thrust and power loading, Pareto solutions can be derived (see Fig. 24). To achieve the maximum aerodynamic efficiency when the required thrust coefficient is 0.0162, the design parameter could be set to $(Z, \phi, \Delta\theta) = (0.25D, -46.1^\circ, 4.0^\circ)$. Similarly, when the required thrust coefficient is 0.0151, the highest power loading can be obtained by

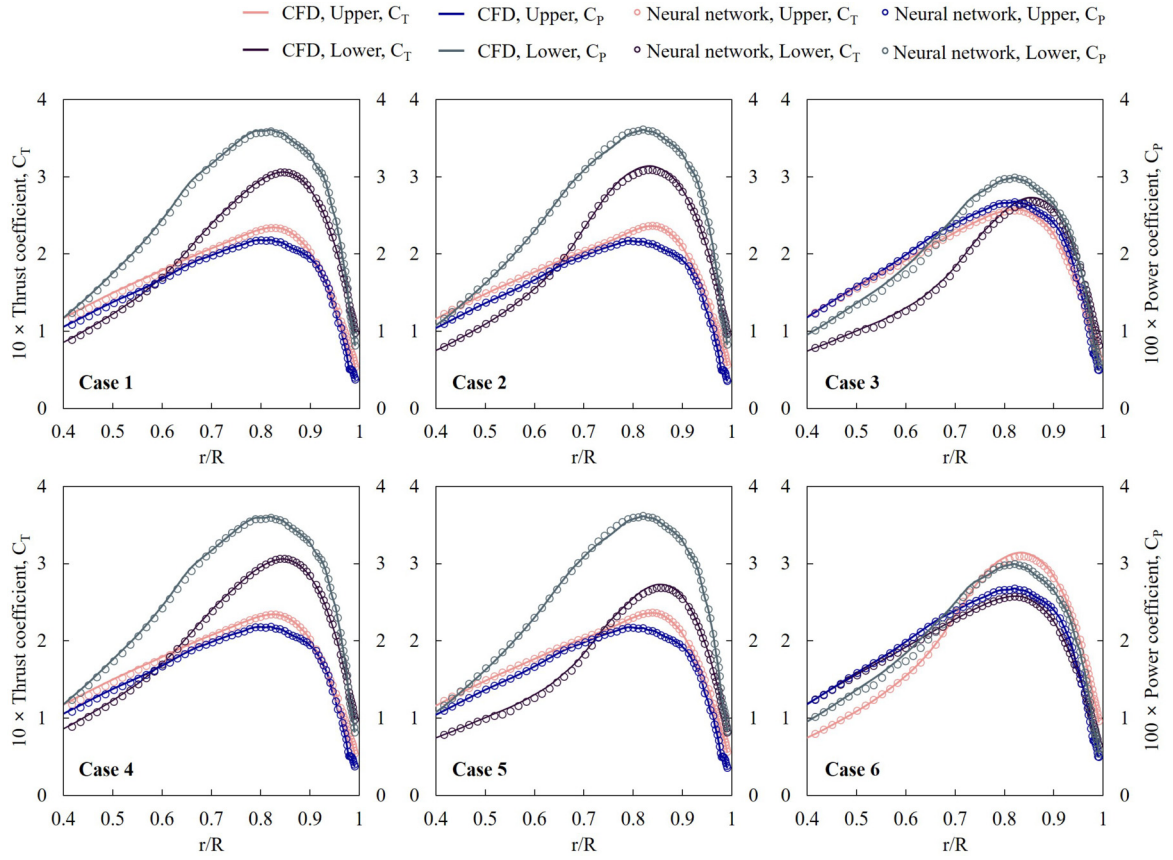


Fig. 22. Validation of the trained two-stage framework using test dataset.

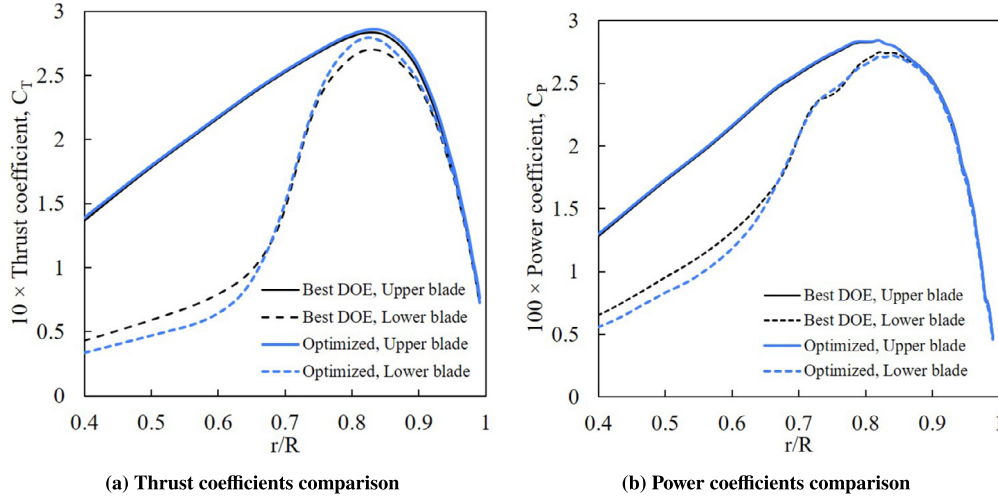


Fig. 23. Comparison of sectional aerodynamic coefficients of the optimized design and best DOE case.

setting the design parameter to $(Z, \phi, \Delta\theta) = (0.36D, 50.1^\circ, 0.0^\circ)$. In conclusion, these results demonstrate that one can calculate the required thrust for an aircraft based upon the payload and operational profile, and then change the design variables to achieve the highest aerodynamic efficiency.

The design variables of Pareto solutions are also described in Fig. 24. The design rules for stacked rotors can be achieved by utilizing the underlying physics described above. Considering the combination of the index angle and the stacked distance of the Pareto solutions, it can be seen that they generally have the BVI on the lower blades. In the region where the thrust coefficient is 0.151~0.158, the stacked distance is 0.25D~0.37D, and the index

angle is $25^\circ \sim 50^\circ$. As shown in Fig. 15, BVI occurs in the lower blade of the configurations. These results indicate that the BVI in the lower blade plays a significant role in steady aerodynamic performance. It should also be noted that all Pareto solutions have neither very small nor very large stacked distances. Accordingly, the inflow effect of the upper and lower blades must be optimized for optimal aerodynamic efficiency.

Thrust and pitch angle differences have a near linear relationship. As $\Delta\theta$ increases, the total thrust also increases because the increased thrust of the lower blade is greater than the decreased thrust of the upper blade. However, when the thrust coefficient exceeds 0.016 (or the power loading is about 9.4), $\Delta\theta$ reaches a

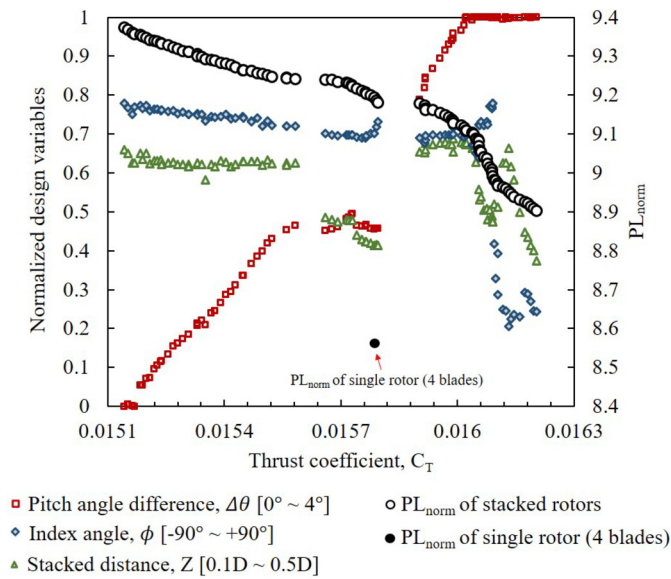


Fig. 24. Pareto solutions of multi-objective optimization problem (maximize PL_{norm} and C_T).

maximum value and remains as a constant value, 1. It indicates that to achieve the higher power loading, the upper bound of the design space in terms of $\Delta\theta$ should be expanded. Since its upper bound is adopted as 4° in this study to prevent the stall of the lower blade due to high $\Delta\theta$. This inspection can also be supported by the fact that the index angle and stacked distance fluctuate in this region without a clear trend to be analyzed.

7. Conclusion

Stacked rotors with various index angles, stacked distances, and pitch angle differences have been numerically investigated under hovering conditions. The main objective of this study is to disclose and comprehend the fundamental physics of the flowfields generated by stacked rotors. Accordingly, the inflow and BVI effects were identified as the two dominant steady effects. The unsteady effect associated with BVI was also examined. With the design optimization based on the two-stage deep neural networks, optimal aerodynamic efficiency can be achieved by taking full advantage of the inflow and BVI effects. Moreover, a design rule was extracted to attain maximum efficiency according to the target thrust. Following is a summary of the findings of this study:

- 1) Based on the analysis of three design parameters and a large design space, the aerodynamic performance of all stacked rotor configurations was at least 1.35% to 8.65% greater than that of a single rotor with four blades. These effects result from the increase in the distance between blades through stacking. The tip vortex effect on aerodynamic performance becomes relatively small, which is related to the increase in the overall effective angle of attack. However, the aerodynamic performance of all lower blades, regardless of their configuration, is lower than a single rotor with two blades. Similarly, upper blades with the same pitch angle as the two-bladed single rotor also exhibit poor aerodynamic performance. These results can be explained as inflow and BVI effects.
- 2) The interaction between the upper and lower blades of the stacked rotor increases the inflow. The increase in inflow will result in a decrease in the effective angle of attack as well as a reduction in aerodynamic efficiency. Various mechanisms contribute to the inflow of the upper and lower blades. Firstly, the wake of the upper blade is absorbed by the lower blade. This

absorption accelerates the inflow of the upper blade. As the stacked distance increases, this effect weakens. On the other hand, in the case of the lower blade, the wake from the upper blade directly increases the inflow of the lower blade, except in the blade tip region ($r/R \geq 0.9$). Consequently, the aerodynamic performance of the lower blade is reduced.

- 3) The interaction between the lower blade and the tip vortex of the upper blade can optimize aerodynamic performance. Due to the upwash created by the vortex, local power loading is increased in the outer region of the blade. There is some downwash generated in the inner region, which reduces the local power loading, but this has a relatively small impact on aerodynamic performance due to the relatively low freestream velocity.
- 4) Unsteady analyses demonstrated that BVI in stacked rotors has little effect on unsteady aerodynamic loads. In the case of the stacked rotor geometry for both with and without BVI, there was no significant difference in the thrust fluctuations. Further, the peak-to-peak thrust of the stacked rotor is approximately 60% less than that of the counter-rotating coaxial rotor. It is therefore reasonable to assume that the tonal noise of the stacked rotor will be much lower than that of the counter-rotating coaxial rotor.
- 5) Two-stage neural networks were employed to optimize the sectional thrust and power distribution. The identified optimal configurations were similar to the best DOE configurations. The optimized case maximizes the local power loading of the lower blade. The two-stage neural networks presented in this study allow us to find the optimal point and identify the design parameter that will create the highest aerodynamic efficiency for the given thrust. This will enable the stacked rotor configuration to be modified according to the mission profile and payload to achieve the best aerodynamic efficiency when applied to actual aircraft.

An in-depth study of stacked rotor aerodynamic performance is presented in this paper, which is essential for the design of actual rotors early in the development process. A few limitations exist in this study, such as the absence of consideration of flight conditions other than hovering and the absence of consideration of structures that may exist in real aircraft, such as additional rods or fuselage. In future research, it is necessary to examine a wide variety of flight conditions. Furthermore, aeroacoustic and structural performance must be thoroughly examined to ensure a robust aircraft design.

Declaration of competing interest

The authors certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

Data availability

Data will be made available on request.

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